The Copperbelt University

Kapasa Makasa University Campus

Department of Agriculture and Aquatic Sciences

SESSIONAL EXAMINATIONS 2021

COURSE CODE: NR 350

COURSE NAME: ANIMAL SCIENCE

TIME ALLOWED: THREE (3) HOURS

DATE: 12TH NOVEMBER 2021

INSTRUCTIONS: 1. There are seven questions in total

2. Answer the COMPULSORY question one and any other FOUR questions.

3. There are 100 marks allocated to this paper

4. Each question carries 20 marks

5. Where possible, illustrate your answers

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Question Five

a) Give a detailed outline of the unit structure of the mammalian urinary system. A diagram could be helpful. **(5 marks)**

b) How is the urinary system for mammals different from that for poultry? **(15 marks)**

Question Six

a) Explain the basic difference between the meiotic and mitotic cell division and why they occur. **(6 marks)**

b) Give a detailed explanation of mitotic cell division. **(8 marks)**

c) Discuss how Gonadotrophin releasing hormone (GnRH) functions to regulate spermatogenesis in mature male animals. **(6 marks)**

Question Seven

You are visiting your village after a COVID-19 induced closure of the university. The community is aware that you are studying for a degree in Animal Sciences and they engage you to explain why previously active bulls within the community occasionally express low ***libido***. Give a detailed outline of your explanation to the enthusiastic crowd gathered to listen to you. **(20 marks)**

END OF THE EXAMINATION, I WISH YOU ALL THE BEST

- low level of testosterone
- Environmental factor
- competition from other bulls
- diet
- lack exercise
- too not getting enough light

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Question One (COMPULSORY)

a) What are the feeding features of note when incorporating soyabeans, cotton seed and molasses in ruminant animal diets? - Level of inclusion based on animal/species/product stage or type

(4 marks)

b) Mr Sansakuwa yaBidansa is a prominent pig farmer in Nalolo district and he is targeting the festive period to supply 800 porkers on the market from his recently weaned piglets. If the piglets have an average daily feed intake of 1800g/piglet/day, work out a 21% crude protein grower ration to be fed over a 90 day period from the following ingredients whose crude protein content was determined through proximate analysis and results attached:

7% CP low grade maize, 14% CP soyabean cake, 5.4% CP molasses, 60% CP fishmeal, 50% CP meat and bone meal and 9.1% CP sorghum meal. Mr yaBindasa has 4 bags of low grade maize that he has salvaged from his storage room each weighing 88.2kg, 90kg, 72kg and 85.5kg, respectively. He has also obtained 16 boxes of meat and bone-meal each weighing 1.95kg from the local Agro supply shop.

1800 = 1.8kg
800 = x
x = 1.8 x 800
1440 kg/day for 800 piglets

1000g = 1kg
1800g = x
x = 1800/1000
x = 1.8 kg/pig/day
1.8x

Note: Work out the ration to two decimal places.

(16 marks)

1440 kg = 1 day
x = 90
129.600 kg / 800 piglets / 90 days

-energy = 7.17%. protein = 51.33%.

Question Two

You have been invited to give a keynote speech at the local District Livestock Farmers' quarterly meeting and you decide to enlighten the gathering on the key points to consider when deciding which ingredients to use in formulating a ration for different animal species on the farm. What will be the contents of your presentation?

(20 marks)

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SESSIONAL EXAMINATIONS 2020

COURSE CODE: NR 350

COURSE NAME: ANIMAL SCIENCE

TIME ALLOWED: THREE (3) HOURS

DATE: 16TH OCTOBER 2020

INSTRUCTIONS: 1. Answer the COMPULSORY QUESTION ONE

2. Answer any other 4 questions

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Question Three

Explain the function of the following feed additives in animal production:

a) Growth hormones (5 marks)

b) Mineral and vitamin additives (5 marks)

c) Antibiotics (5 marks)

d) Amino acid concentrates (5 marks)

Question Four

a) Give a detailed outline of the factors that affect the nutritive value of herbage (both planted and natural grassland/rangeland). (10 marks)

- leaching
- age
- water
- oxidation
- time spent on wilting
- species type

b) What are the pros and cons of including legumes in a mixed-species pasture stand? (10 marks)

pros
- highly recover rate & fast growth.
- nitrogen fixer
- drought resistance
- high levels of protein & less loss of nutrients with age increase
- less tannin helps to prevent bloat & gases

Question Five

In recent years, there have been arguments for and against the consumption of animal products and by-products by human beings. Discuss. (20 marks)

① Animals provide ½ energy needed by human
② provide ⅓ protein
③ Provide ... Swahili ... B12 not found ...

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Question Six

The circulation of blood in the animal body is in such a way that oxygenated blood and deoxygenated blood do not mix for an animal with a well functioning cardiac system. Discuss.

(20 marks)

Question Seven

Compare and contrast the digestive anatomy of poultry to that of ruminant animals.

(20 marks)

END OF THE EXAMINATION, I WISH YOU ALL THE BEST

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Question One (COMPULSORY)

a) Compare and contrast the female reproductive systems of mammals and the avian species with emphasis on the key features that are similar and those that are dissimilar.

(10 marks)

b) Explain the term **cryptorchidism** in detail and mention its pros and cons.

(10 marks)

Question Two

Write some short but concise notes on the following terms as they apply to reproduction in farm animals:

a) Polytochous animals **(5 marks)**

b) Polyestrous animals **(5 marks)**

c) Monoestrous animals **(5 marks)**

d) Monotochous animals **(5 marks)**

Question Three

With two examples each, give a detailed sequence of events to account for the differences between animals that are called spontaneous ovulators and those called reflex ovulators.

(20 marks)

Question Four

Blood movement in farm animals occurs in what is called pulmonary and systemic circulation. Discuss.

(20 marks)

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Question One (COMPULSORY QUESTION)

The following weaned steers have been targeted for a Christmas party for Kapasa Makasa University students to be held in seven weeks time from today 12th November 2021. The following are the individual weights of the animals (145kg, 115kg, 138kg, 120kg and 124kg). The attached Table 8 indicates the nutrient requirements of beef cattle at different weight bands and planned average daily weight gain.

b) If the animals are targeted to gain 0.9kg/animal/day, work out the amount of feed they will consume over the seven week feeding period from the following ingredients: Maize (CP 9%), Maize bran (CP 6.5%), Soyabeans (CP 46%), Lucerne and Rhodes grass hay (CP 17%).

(18 marks)

a) What will be the average weight of the animals after the feeding period? **(2 marks)**

Question Two

Discuss the different perceptions that surround the keeping of cattle in different societies worldwide.

(20 marks)

Question Three

With three examples each, give a detailed explanation of the following terms as they relate to animal health in The Republic of Zambia:

(10 marks)

a) Zoonotic diseases

(10 marks)

b) Notifiable diseases

Question Four

The adverse tropical climate generally pauses a challenge when it comes to livestock production in the third world countries. Discuss how the environment under which livestock are being reared can be modified to minimize the adverse effects of the harsh tropical conditions.

(20 marks)

Zoonotic
- sleeping sickness
- chicken pox

Notifiable
- Anthrax
- African swine fever
- mouth and foot

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Question Five

Give a detailed description of synergistic hormonal and antagonistic hormonal associations in the animal body with three examples each.

Question Six (20 marks)

An artificial insemination technician has recorded the following as successful inseminations that have been conducted at Kabulonga Farms in Chadiza district.

DATE	Cow ID
03/11/2011	KBF 25
09/11/2021	KBF 33
16/11/2021	KBF 45
22/11/2021	KBF 61
27/11/2021	KBF 73
30/11/2021	KBF 88

a) What are the disadvantages of this insemination schedule? (4 marks)

b) Work out the projected date that each cow will not return to heat. (6 marks)

c) What is the average expected date of parturition for these breeding animals? (10 marks)

Question Seven

There have been gigantic strides made in the livestock development sector in Zambia where superior and exotic breeds of livestock have been introduced in the indigenous gene pool to improve performance and productivity in general. Discuss the considerations to be made when introducing exotic traits in: (10 marks)

a) Poultry Improvement Programmes (10 marks)

b) Beef Improvement Programmes

END OF THE EXAMINATION – I WISH YOU ALL THE BEST

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